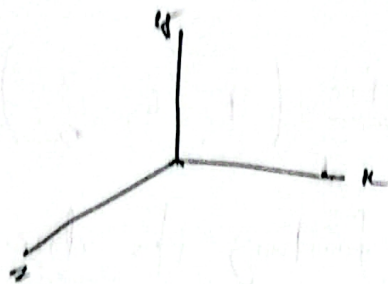


Lecture # 4: Elements of three dimensional geometry ①



The planes $x=0$, $y=0$ and $z=0$ divide space into eight octants.

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be two points in plane then distance between them is

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Distance formula in 3D

Let $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ be distinct points in space such that PQ is not parallel to one of the coordinate axes. Then

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Mid point of two points

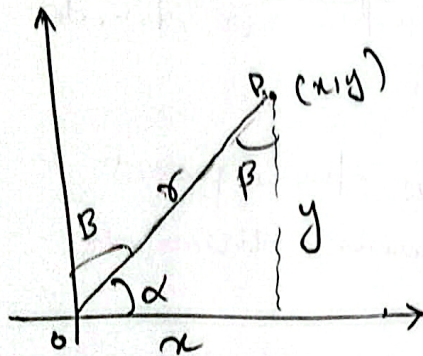
If R is the middle point of the segment PQ , the coordinates of the middle point are

$$x = \frac{x_1 + x_2}{2}, \quad y = \frac{y_1 + y_2}{2}, \quad z = \frac{z_1 + z_2}{2}$$

Let us consider two points $A(3, 2, 1)$ and $B(6, 10, -1)$ then

the mid point C of AB has coordinates
 $\left(\frac{3+6}{2}, \frac{2+10}{2}, \frac{1-1}{2}\right)$ i.e. $\left(\frac{9}{2}, 6, \frac{0}{2}\right)$

Given a point finding its direction cosines

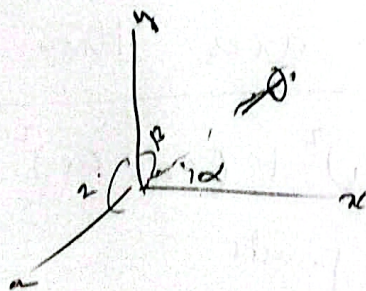


$$\cos \alpha = \frac{x}{r}$$

$$\cos \beta = \frac{y}{r}$$

Direction Angles

The direction angles α, β, γ of a line is defined as
 α = Angle b/w line and the positive x -axis



β = Angle between line and positive y -axis.

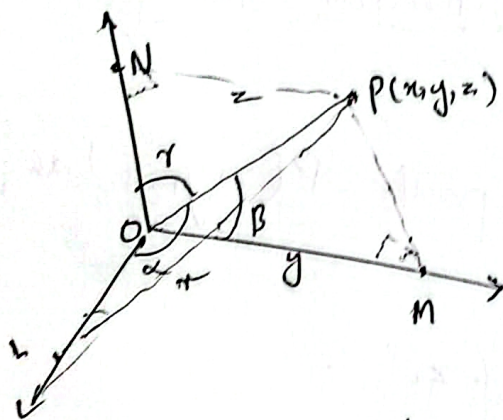
γ = Angle between line and the positive z -axis.

By definition, each of these angles lies between 0 and π .

Direction Ratios

- cosines of direction angles are called direction cosines.
- Any multiple of direction cosines are called direction numbers or direction ratios of the line L .

Direction angles of a line



Direction cosines

$$\cos \alpha = \frac{x}{OP} = \frac{x}{\sqrt{x^2 + y^2 + z^2}}$$

$$\cos \beta = \frac{y}{OP} = \frac{y}{\sqrt{x^2 + y^2 + z^2}}$$

Similarly

$$\cos \gamma = \frac{z}{OP} = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = \frac{x^2}{(OP)^2} + \frac{y^2}{(OP)^2} + \frac{z^2}{(OP)^2}$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = \frac{x^2 + y^2 + z^2}{OP^2}$$

where

$$(OP)^2 = x^2 + y^2 + z^2$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

Direction cosines and direction ratios of line joining two points

For a line joining two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ the direction ratios are

$$x_2 - x_1, y_2 - y_1, z_2 - z_1$$

and the direction cosines are

$$\frac{x_2 - x_1}{|PQ|}, \frac{y_2 - y_1}{|PQ|}, \frac{z_2 - z_1}{|PQ|}$$

Example: For a line joining two points $P(1, 3, 2)$ and $Q(7, -2, 3)$ the direction ratios are:

$$7 - 1 = 6$$

$$-2 - 3 = -5$$

$$3 - 2 = 1$$

and direction cosines are

$$\frac{6}{\sqrt{62}}, \frac{-5}{\sqrt{62}}, \frac{1}{\sqrt{62}}$$

Graph in 2D space

(5)

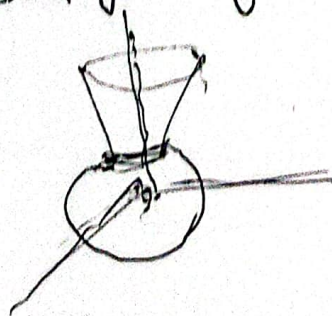
In 2D space the graph of an equation relating the variables x and y is the set of all point (x, y) whose coordinate satisfy the equation. Usually such graphs are curves.

Graph in 3D space

In 3D space the graph of an equation relating the variables x, y and z is the set of all points (x, y, z) whose coordinate satisfy the equation, usually such graphs are surfaces.

Intersection of two surfaces.

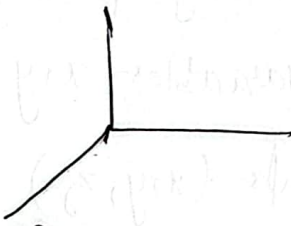
- Intersection of two surfaces is curve in three dimensional space
- It is the reason that a curve in three dimension space is represented by two equations representing the intersecting surfaces.



Intersection of two planes

If two planes are not parallel, then they intersect and their intersection is a straight line.

Thus, two non-parallel planes represent a straight line given by two simultaneous linear equations in x, y and z and are known as non-symmetric form of equations of a straight line.



Eight octants in 3D:

Region

Description

Equation

xy

consists of all points of the form $(x, y, 0)$

$$z=0$$

xz

consist of all points of the form $(x, 0, z)$

$$y=0$$

yz

consist of all points of the form $(0, y, z)$

$$x=0$$

Position
x-axis

Description
consists of all points
of the form $(x, y, 0)$

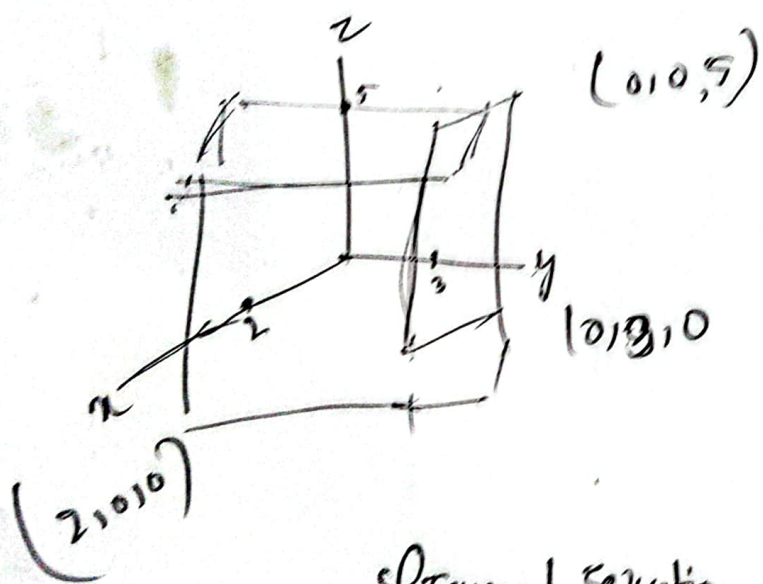
Equation
 $z=0, y=0$

y-axis consists of all points of the
form $(0, y, 0)$

$z=0, x=0$

z-axis consists of all points of the
form $(0, 0, z)$

$x=0, y=0$



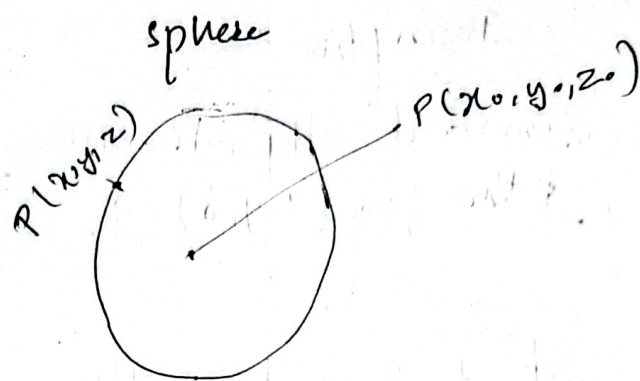
General equation of the plane in 3D

any equation of the form:

$$ax + by + cz + d = 0$$

where a, b, c, d are real numbers represent plane.

⑧



→ The standard equation of the sphere
radius a centered at (x_0, y_0, z_0) is
$$(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2 = a^2$$